

Foundations: Itô Calculus and the SDE Framework

`hasklib` mathematical companion

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July 18, 2026

Abstract

This note collects the stochastic-calculus results that the `stochastic/` layer of `hasklib` relies on: the probability-space setup, Brownian motion, the general Itô SDE $dX_t = a(t, X_t)dt + b(t, X_t)dW_t$ (encoded by the `StochasticProcess` interface), Itô's lemma, and quadratic variation. Downstream files (`gbm.tex`, `ou.tex`, `cir.tex`, `schemes.tex`) cite the results established here. Standard measure-theoretic probability is assumed; see Shreve (2004) or Williams (1991) for construction.

1 Probability space and Brownian motion

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. The filtration $\{\mathcal{F}_t\}$ represents the information available up to time t .

Definition 1 (Brownian motion). A standard Brownian motion $\{W_t\}_{t \geq 0}$ is a process with

1. $W_0 = 0$ almost surely;
2. Independent increments: for $s < t < u$, $W_u - W_t$ is independent of $W_t - W_s$;
3. Gaussian increments: $W_t - W_s \sim \mathcal{N}(0, t - s)$;
4. Continuous sample paths.

Remark 1. In code, property (3) is what `NormalRng::draw()` supplies: a step of length Δt uses $W_{t+\Delta t} - W_t = \sqrt{\Delta t} Z$ with $Z \sim \mathcal{N}(0, 1)$. This substitution is the single bridge between the continuous theory here and the discrete simulation code.

Although the paths are continuous, they are extremely rough. Fix t and consider the difference quotient over a step $h > 0$. By the Gaussian-increment property, $W_{t+h} - W_t \sim \mathcal{N}(0, h)$, equivalently $\sqrt{h} Z$ with $Z \sim \mathcal{N}(0, 1)$, so

$$\frac{W_{t+h} - W_t}{h} \stackrel{d}{=} \frac{1}{\sqrt{h}} Z,$$

whose standard deviation $1/\sqrt{h}$ diverges as $h \downarrow 0$. Informally, the path moves by order $\sqrt{h}$ over a time step of order h , so its instantaneous slope is of order $1/\sqrt{h} \rightarrow \infty$ and the quotient converges to no finite limit. This heuristic is made rigorous by the Paley–Wiener–Zygmund theorem (Paley, Wiener & Zygmund 1933): with probability one, $t \mapsto W_t$ is nowhere differentiable.

Equivalently, the path has infinite first-order (total) variation on every interval, so integrals $\int f dW_s$ cannot be defined pathwise as Riemann–Stieltjes integrals. Its *quadratic* variation, however, is finite — the content of section 2 — and it is precisely this finite second-order variation that the Itô integral and Itô's lemma are built on.

2 Quadratic variation

The property that distinguishes stochastic from ordinary calculus is that Brownian motion accumulates *quadratic* variation at a nonzero rate.

Proposition 1 (Quadratic variation of Brownian motion). *For a standard Brownian motion, the quadratic variation over $[0, T]$ satisfies*

$$\langle W \rangle_T = \lim_{\|\Pi\| \rightarrow 0} \sum_k (W_{t_{k+1}} - W_{t_k})^2 = T \quad \text{in } L^2,$$

where Π is a partition of $[0, T]$. Heuristically this is written

$$(dW_t)^2 = dt.$$

Proof. Write the partition as $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$ and abbreviate

$$\Delta t_k = t_{k+1} - t_k, \quad \Delta W_k = W_{t_{k+1}} - W_{t_k}, \quad S_\Pi = \sum_{k=0}^{n-1} (\Delta W_k)^2.$$

“Convergence in L^2 ” means the mean-square error vanishes: $\mathbb{E}[(S_\Pi - T)^2] \rightarrow 0$ as the mesh $\|\Pi\| := \max_k \Delta t_k \rightarrow 0$. We use only two facts about the increments, both directly from the definition of Brownian motion — each $\Delta W_k \sim \mathcal{N}(0, \Delta t_k)$, and increments over disjoint intervals are independent. Nothing measure-theoretic enters beyond these.

Step 1 — the mean is exactly T . Since $\mathbb{E}[\Delta W_k] = 0$, the second moment is the variance:

$$\mathbb{E}[(\Delta W_k)^2] = \text{Var}(\Delta W_k) = \Delta t_k.$$

Summing over k ,

$$\mathbb{E}[S_\Pi] = \sum_k \mathbb{E}[(\Delta W_k)^2] = \sum_k \Delta t_k = T,$$

which holds for *every* partition, not merely in the limit. Because the mean is exactly T , the mean-square error *is* the variance:

$$\mathbb{E}[(S_\Pi - T)^2] = \text{Var}(S_\Pi).$$

So the whole statement reduces to showing $\text{Var}(S_\Pi) \rightarrow 0$.

Step 2 — variance of a single term. Normalise the increment as $\Delta W_k = \sqrt{\Delta t_k} Z$ with $Z \sim \mathcal{N}(0, 1)$, so $(\Delta W_k)^2 = \Delta t_k Z^2$ and

$$\text{Var}((\Delta W_k)^2) = (\Delta t_k)^2 \text{Var}(Z^2).$$

For a standard normal, $\mathbb{E}[Z^2] = 1$ and $\mathbb{E}[Z^4] = 3$, hence $\text{Var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = 3 - 1 = 2$. (The fourth moment is one Gaussian integration by parts: with the density ϕ satisfying $\phi'(z) = -z\phi(z)$, integrating $\int z^4 \phi dz$ by parts gives $\mathbb{E}[Z^4] = 3\mathbb{E}[Z^2] = 3$.) Therefore

$$\text{Var}((\Delta W_k)^2) = 2(\Delta t_k)^2.$$

Step 3 — variance of the sum, then let the mesh shrink. The increments ΔW_k are independent, so the squared increments $(\Delta W_k)^2$ are independent too, and the variance of a sum of independent variables is the sum of the variances:

$$\text{Var}(S_\Pi) = \sum_k \text{Var}((\Delta W_k)^2) = 2 \sum_k (\Delta t_k)^2.$$

Bound one factor of Δt_k in each term by the mesh:

$$\sum_k (\Delta t_k)^2 \leq \left(\max_k \Delta t_k \right) \sum_k \Delta t_k = \|\Pi\| T.$$

Combining,

$$\mathbb{E}[(S_\Pi - T)^2] = \text{Var}(S_\Pi) \leq 2 \|\Pi\| T \xrightarrow{\|\Pi\| \rightarrow 0} 0,$$

which is exactly the claimed L^2 convergence. $\square$

Remark 2. The mean computation ($\mathbb{E}[S_\Pi] = T$) is what makes $(dW_t)^2 = dt$ *plausible*; the variance computation is what makes it *usable*. Per step the squared increment $(\Delta W_k)^2$ still fluctuates — its standard deviation $\sqrt{2} \Delta t_k$ is the same order as its mean — but those fluctuations are independent across steps, so summing averages them out at rate $\|\Pi\|$. Inside a sum (equivalently, an integral) the random $(\Delta W_k)^2$ may therefore be replaced by the deterministic Δt_k . That replacement is the entire content of the shorthand $(dW_t)^2 = dt$ used in the next section.

3 The Itô stochastic differential equation

Definition 2 (Itô SDE). An Itô process X_t solves the stochastic differential equation

$$dX_t = a(t, X_t) dt + b(t, X_t) dW_t, \quad (1)$$

interpreted as the integral equation

$$X_t = X_0 + \int_0^t a(s, X_s) ds + \int_0^t b(s, X_s) dW_s,$$

where the second integral is an Itô integral. We call a the *drift* and b the *diffusion* coefficient.

Remark 3 (Correspondence with the code). Equation (1) is exactly what the `StochasticProcess` base class encodes: `drift(t,x)` returns $a(t,x)$ and `diffusion(t,x)` returns $b(t,x)$. Every concrete process (GBM, OU, CIR) is a choice of these two functions.

Assumption 1 (Existence and uniqueness). We assume a and b satisfy the standard Lipschitz and linear-growth conditions guaranteeing a unique strong solution to (1); see Øksendal (2003), Thm. 5.2.1. (*Cited, not proved — the library never depends on the proof, only on the conclusion.*)

4 Itô's lemma

Itô's lemma is the chain rule for functions of an Itô process, and it is the engine behind every closed-form solution used in `hasklib` (the GBM exact sampler, the OU solution).

Theorem 1 (Itô's lemma, one dimension). *Let X_t solve (1) and let $f(t, x) \in C^{1,2}$. Then*

$$df(t, X_t) = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) dt + b f_x dW_t, \quad (2)$$

where subscripts denote partial derivatives evaluated at (t, X_t) .

Heuristic derivation. This is the standard second-order Taylor argument. It is not a proof in the strict sense — a rigorous version requires the construction of the Itô integral (Shreve 2004, Ch. 4), which is outside the scope of this note — but it exposes exactly where the extra term comes from, and it is all the library needs.

The one idea is this: because dX carries a Brownian term of size $\sqrt{dt}$, a contribution that ordinary calculus discards as “second order” is in fact first order and must be kept.

Orders of magnitude. Over a step of length dt the increment dW_t has typical size $\sqrt{dt}$ (it is $\mathcal{N}(0, dt)$). Ranking the elementary products by their power of dt :

$$dt = O(dt), \quad dW_t = O(dt^{1/2}), \quad (dW_t)^2 = O(dt), \quad dt dW_t = O(dt^{3/2}), \quad (dt)^2 = O(dt^2).$$

Discard everything of order smaller than dt , and use proposition 1 to replace $(dW_t)^2$ by its in-sum limit dt . This gives the *Itô multiplication table*

$$(dW_t)^2 = dt, \quad dt dW_t = 0, \quad (dt)^2 = 0. \quad (3)$$

The last two entries are not literally zero; they are negligible next to dt and vanish in the mesh limit for exactly the reason the fluctuations did in section 2.

Taylor expansion. Expand $f(t + dt, X_t + dX)$ about (t, X_t) to second order:

$$df = f_t dt + f_x dX + \frac{1}{2} f_{tt} (dt)^2 + f_{tx} dt dX + \frac{1}{2} f_{xx} (dX)^2 + \dots$$

Substitute the SDE $dX = a dt + b dW_t$ and square:

$$(dX)^2 = a^2 (dt)^2 + 2ab dt dW_t + b^2 (dW_t)^2.$$

Apply (3). In $(dX)^2$ the first two terms die and only $b^2 (dW_t)^2 = b^2 dt$ survives:

$$(dX)^2 = b^2 dt.$$

The two remaining second-order terms of the expansion vanish for the same reason:

$$\frac{1}{2} f_{tt} (dt)^2 = 0, \quad f_{tx} dt dX = f_{tx} (a (dt)^2 + b dt dW_t) = 0.$$

What remains is

$$df = f_t dt + f_x dX + \frac{1}{2} f_{xx} b^2 dt.$$

Substitute $dX = a dt + b dW_t$ once more and collect the dt and dW_t parts:

$$df = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) dt + b f_x dW_t,$$

which is (2). □

Remark 4. Had X been a smooth (finite-variation) path, $(dX)^2$ would have been genuinely $O((dt)^2)$ and dropped out, leaving the ordinary chain rule $df = f_t dt + f_x dX$. The whole gap between stochastic and ordinary calculus is the single surviving term $\frac{1}{2} b^2 f_{xx} dt$, and it survives only because $(dW_t)^2 = dt$ rather than 0 — i.e. because Brownian motion has nonzero quadratic variation. Every closed-form result in `hasklib` that “corrects a drift by half a variance” (the $-\frac{1}{2}\sigma^2$ in the GBM log-solution, the Itô term in the OU solution) is exactly this one term.

Remark 5. The single downstream consequence to keep in view: applying theorem 1 with $f(t, x) = \ln x$ to geometric Brownian motion linearises the SDE and yields the exact terminal law sampled by `GBM::sample_terminal`. That derivation is carried out in `gbm.tex`.

5 What the following files derive

Each companion file starts from (1) with a specific (a, b) and uses theorem 1 to produce results the code depends on:

- `gbm.tex`: $a = \mu x$, $b = \sigma x \Rightarrow$ exact solution and moments (verified in `test_stochastic.cpp`).
- `ou.tex`: $a = \kappa(\theta - x)$, $b = \sigma \Rightarrow$ Gaussian solution, mean/variance, stationary law.
- `cir.tex`: $a = \kappa(\theta - x)$, $b = \sigma\sqrt{x} \Rightarrow$ moments, Feller condition, motivation for full truncation.
- `schemes.tex`: discretisation of (1) — Euler–Maruyama (strong order 0.5) and Milstein (strong order 1.0) via stochastic Taylor expansion (Kloeden & Platen 1992).

References

- Kloeden, P. E. & Platen, E. (1992), *Numerical Solution of Stochastic Differential Equations*, Vol. 23 of *Applications of Mathematics*, Springer, Berlin.
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- Williams, D. (1991), *Probability with Martingales*, Cambridge University Press, Cambridge.